

SECTION 15063-A
STAINLESS STEEL PIPE, FITTINGS, AND VALVES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions.

PART 2 - PRODUCTS

2.1 304 STAINLESS STEEL PIPE

A. Pipe:

- 1. Seamless or welded Stainless steel, Schedule 20, Grade TP304. Pipe shall conform to ASTM A312.

B. Fittings:

- 1. Stainless steel, Schedule 20, butt weld, Grade F 304 . Fittings shall conform to ASTM A182.

C. Flanges:

- 1. Class 150 Stainless steel, weld neck, raised face, Grade F304. Fittings shall conform to ASTM A1 82.

D. Gaskets:

- 1. Standard pre-cut non-asbestos Garlock Blue-Gard Style 3200 1/16" thick gasket, conforming to the full gasket face of the flange.

E. Valves:

- 1. 2 inches and smaller - Stainless steel Ball valve.
- 2. 2-1/2 inches and larger - Stainless steel wafer type.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Ream pipes and tubes to full inside diameter. Clean off scale and dirt, inside and outside, before assembly.

Remove welding slag or other foreign material from piping.

3.2 CONNECTIONS

- A. Screw joint steel piping up to and including 2". Weld piping 2-1/2" and larger, including branch connections.
- B. Use full-cut standard taper pipe threads. Make up joints using Teflon tape or nontoxic joint compound as related to the piping system involved. Apply to male threads only.
- C. Use main-sized "Weld-O-Let" or 'Thread-O-Let" male branch connections or 'Stub-in' in steel piping if main is at least one pipe size larger than the branch for up to 6' mains and if main is at least two pipe sizes larger than branch for 8' and larger mains. Do not project branch pipes inside the main pipe.
- D. Make connections to equipment and branch mains with unions or flanges.

3.3 ROUTE AND GRADES

- A. Route piping in orderly manner and maintain proper grades. Install to conserve headroom and interfere as little as possible with use of space. Run exposed piping parallel to walls. Group piping whenever practical at common elevations. Install concealed pipes close to building structure to keep furring to a minimum.
- B. Slope water piping 1" in 40' and arrange to drain at low points.
- C. On closed systems, low points shall be provided with 3/4" drain valves and hose nipples. Provide manual air vents at high points.
- D. Make reductions in water pipes with eccentric reducing fittings installed to provide drainage and venting.
- E. Install piping to allow for expansion and contraction without stressing pipe or connected equipment.
- F. Provide clearance for access to valves, air vents, drains and unions.

3.4 CLEANING AND TESTING

- A. All testing shall be done in the presence of the Owner's representative.
- B. Do not test relief valves, pressure-reducing valves, valves, or equipment beyond its rated capacity.
- C. Process Cooling Water Piping System: Flush pipe free of gross dirt and debris with clear water. Pressure test with water at working pressure by 1.5 factor. Visually inspect joints for leaks, repair or replace as required and retest. The clean room contractor shall clean up Y-strainers and heat exchangers after pipe flush.

3.5 PIPING SYSTEM LEAK REPAIR

- A. Leaks which occur during the period of warranty shall be repaired, including any building damage or refinishing costs, at no cost to Owner.
- B. All defects which develop under tests shall be repaired promptly and the tests shall be repeated. No caulking of screwed joints, cracks, or holes will be permitted.
- C. Minor leaks in welded joints may be chipped out and rewelded. A general sweating of a welded joint may be cause for rejection of the entire joint.
- D. Leaks in threaded pipe shall be repaired by replacing the thread or the fitting or both. Any replacement piece shall be the same length as the piece removed.

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